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Material Constants of Anisotropic Poroelasticity and Its Impacts on Shale Gas Production

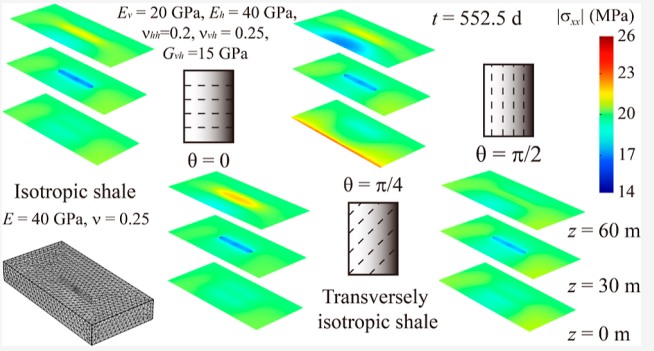
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ABSTRACT: While the production of shale gas is always accompanied by stratum deformation, previous studies have commonly assumed isotropy to simplify the modeling process despite substantial experimental evidence supporting the anisotropy of these formations. This work contributes to both theoretical and practical aspects of anisotropy. For the theoretical aspect, a novel mixture theory approach is used to explore the poromechanical constants of anisotropic poroelasticity. By incorporating new elements in the modeling of intrinsic solid density and unjacketed frame deformation, we are able to establish a complete set of parameters considering micro-inhomogeneity and micro-anisotropy in a macroscopically anisotropic porous medium like shale. The bounds on the material constants are established, and their relationship with the intrinsic material constants of poroelasticity proposed by Cheng (2021) is discussed. In a practical context, we examine the impacts of anisotropy in a 3D shale gas reservoir considering both elastic and permeability anisotropy. For elastic anisotropy, compared with the isotropic counterpart, we identify significant dissimilarities in the horizontal stress parallel to the wellbore (σ_{xx} here) between assumptions of isotropy and transverse isotropy. For permeability anisotropy with an inclined plane of isotropy, the gas production is largely inhibited, and we could observe the stress distribution reorientation. To conclude, isotropic models are unable to reproduce the stress changes predicted by the anisotropic models, and this research contributes to a better understanding of anisotropy in shale gas reservoirs.



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1. INTRODUCTION

Shale gas has emerged as a crucial energy source in recent years due to its abundance, low cost, and reduced carbon footprint compared to other fossil fuels.[1] However, the original shale formations are often hundreds or thousands of feet below the surface and are characterized by low permeability,[1,2] which makes it difficult for gas to flow through the rock and reach the wellbore. Recently, thanks to the advancements in horizontal well drilling and hydraulic fracturing technologies,[3,4] the recovery of natural gas from previously inaccessible shale formations has entered the stage of commercial production. These innovative techniques have revolutionized the production process, unlocking vast reserves of natural gas that were previously deemed uneconomical.[5] To optimize the production process and gain a deeper understanding of the underlying mechanisms, researchers have turned to numerical simulation as a means to visualize the behavior of these fractured reservoirs.[6−8] While the production of shale gas is intrinsically linked to the deformation of the surrounding stratum,[1] most studies adopted the isotropic Hooke's law.[9]

However, shale, as one of the most widespread sedimentary formations on Earth, exposed well-defined anisotropy in the form of bedding, lamination, foliation (repetitive layering), fissuring, or jointing.[10,11] The fissures or joints could generate anisotropic and preferential fluid flow paths[2,12] and lead to a certain level of discontinuity, depending on the size of the problem of interest.[10] For example, for a regional problem on the reservoir scale, the associated deformation discontinuity across enormous fissures and joints could be ignored,[7] while they mostly contribute to the fluid flow through the so-called multiple porosity model.[6,13−15] Therefore, in this study, we focus on the so-called intact anisotropy,[10] which means that for any material point in the domain, the stiffness and/or permeability vary with the observing direction.[2,11,16] This anisotropy can have a significant impact on the way that fractures propagate through the rock,[4,17] the direction in which gas flows,[2,18] and the shear failure.[19−22] Therefore, coupled hydro-mechanical simulations should take the anisotropic characteristics into account.[5,23−25]

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To be able to realize the hydro-mechanical numerical simulation, the key element is the theory of poroelasticity, which was pioneered by Maurice A. Biot in 1941.[26] Through more than 80 years of development, the most widely used poroelastic constants (i.e., drained bulk modulus, bulk modulus of the solid grain, porosity, and fluid bulk modulus) always adopt the ideal porous medium assumption,[27] and they are even treated as standards in many commercial software. In such a model, it is assumed that the solid is made of a single isotropic mineral. However, numerous geomaterials, including sedimentary rocks, consist of multiple minerals accompanied by a proportion of soft clay.[28] Even in the case of a comparatively pure mineral, it may contain enclosed voids that are not interconnected with the flow pathways. These pores must be acknowledged as integral components of the solid material, thereby resulting in its inhomogeneous nature.[29] As a result, it is crucial to establish a comprehensive model that does not depend on the assumption of an ideal porous medium. For anisotropic porous medium, Cheng[30] first derived a complete set of poroelastic constants through micromechanics. Later, Zhao and Borja[31] and Zhang and Borja[32] adopted another two different approaches to arrive at the same result under the ideal porous medium assumption, suggesting that the fluid flow equation can be derived in more than one way. Nevertheless, micro-anisotropy and micro-inhomogeneity are not yet incorporated in their approaches. Therefore, this study aims to fill in this gap in anisotropic poroelastic constants.

Following the direct consequences of anisotropy mentioned in the previous paragraphs, in the model application, we focus on the stress evolution in shale gas reservoirs induced by anisotropy as the stress field plays a central role in the dynamics of fracture and failure. Previous work on stress evolution emphasizes the impacts of complex hydraulic fracture geometry[23] and natural fracture networks such as the inclination angle, permeability, and connectivity to hydraulic fractures,[7] while the reservoir elastic anisotropy is not considered. Recently, Tang et al.[5] considered anisotropy and heterogeneity in the modeling of an actual shale gas reservoir in Sichuan Basin, China, and results suggested that transversely isotropic intensity has a considerable effect on the geomechanics evolution and must be considered.[5,24] In their work, an interface (coupling) Python code is provided to link the CMG with Abaqus through the fixed-stress iterative coupling,[14] which is efficient for large-scale engineering problems. However, the adopted porosity updating equation seems to assume a Biot coefficient of one, which motivates a more reasonable porosity updating equation in the case of anisotropy and compressible solid grain. Furthermore, the anisotropy in Tang et al.[5] is limited to the case of vertical transverse isotropy, but the ground movement could cause the phenomenon of fold in geology; thus, the bedding plane orientation should be considered as well.[31]

This work contributes to both the theoretical and practical aspects of anisotropy, and the paper is organized as follows. Section 2 provides all the mathematical details of the proposed approach that considers micro-anisotropy and micro-inhomogeneity.[29] In Section 3, bounds on the material constants of anisotropic poroelasticity are derived based on thermodynamics and elastic stability requirements, and an extension of the "intrinsic" material constants of poroelasticity[33] is also provided. The next two sections focus on the model application. In Section 4, we first verify the implementation of elastic anisotropy in a 3D hydro-mechanical coupling finite element framework. Then, in Section 5, an example of a 3D anisotropic shale gas reservoir with one hydraulic fracture located at the center of the domain is analyzed in detail. Redistributed stress states in complex anisotropic shale gas reservoirs can be efficiently and accurately simulated, which cannot be achieved by most existing simulators that assume isotropy. The conclusions are given in Section 6. The findings of this study contribute to a more comprehensive understanding of gas production in shale gas reservoirs.

2. DERIVATION OF POROELASTIC COEFFICIENTS CONSIDERING ANISOTROPY AND MICRO-INHOMOGENEITY BY MIXTURE THEORY

In this section, we show through some novel treatments in mixture theory for deformable porous media,[31,32,34] the anisotropy and micro-inhomogeneity could be captured in a unified modeling framework, which would be a great alternative to the micromechanical and phenomenological approach[30,33,35] in the literature.

Let us denote V as the representative elementary volume (REV) consisting of a mixture of solid and fluid (gas), which occupy volumes of V_s and V_p, respectively. Obviously, we require

$$V = V_s + V_p \tag{1}$$

The (current) porosity ϕ is defined as

$$\phi = \frac{V_p}{V} \tag{2}$$

Throughout this paper, we would assume infinitesimal deformation and therefore, for the following gradient and divergence operation, we do not need to distinguish between the reference and current configurations. To describe the mass balance of the solid, we first define the partial solid density ρ^s as

$$\rho^s = \rho_s(1 - \phi) \tag{3}$$

where ρ_s is the intrinsic mass density of the solid. Therefore, the physical meaning of ρ^s is actually the ratio of solid mass in REV to the REV volume V.

Assuming no mass exchange between solid and fluid, the mixture theory gives the universal mass conservation form as

$$\frac{d\rho^s}{dt} + \rho^s \nabla \cdot \boldsymbol{v}_s = 0 \tag{4}$$

which is independent of any specific constitutive properties of the solid, and $\boldsymbol{v}_s$ is the solid velocity vector. By using eq 3, we could rewrite eq 4 as

$$\frac{d\phi}{dt} = \frac{1 - \phi}{\rho_s}\frac{d\rho_s}{dt} + (1 - \phi)\nabla \cdot \boldsymbol{v}_s \tag{5}$$

Inspired by Makhnenko and Podladchikov,[36] we introduce two moduli K_s and K_x to relate change in ρ_s to the changes in the intrinsic solid pressure p_s (compressive stress) and fluid pressure p, respectively, which could be expressed as

$$\frac{1}{\rho_s}\frac{d\rho_s}{dt} = \frac{1}{K_s}\frac{dp_s}{dt} + \frac{1}{K_x}\frac{dp}{dt} \tag{6}$$

Here, K_s could be understood as the drained bulk modulus of the composite solid, i.e., when $p = 0$. K_x is an undetermined quantity whose value will be explored in the next section. According to Cheng,[33] when $p_s = p$, we could determine the "effective bulk modulus of the composite solid" $\overline{K}_s$ as

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$$\frac{1}{\bar{K}_s} = \frac{1}{K_s} + \frac{1}{K_x} \tag{7}$$

By adopting eq 6, eq 5 could be recast into

$$\frac{d\phi}{dt} = \frac{1-\phi}{K_s}\frac{dp_s}{dt} + (1-\phi)\nabla\cdot\boldsymbol{v}_s + \frac{1-\phi}{K_x}\frac{dp}{dt} \tag{8}$$

According to the stress partition theorem used in both mixture theory[34] and variational energy principle,[35] the mean total stress σ_t can be decomposed into a solid part p_s and a fluid part p, weighted by their volume fractions $1-\phi$ and ϕ, written as

$$\sigma_t = -[(1-\phi)p_s + \phi p] \tag{9}$$

where the minus sign indicates a change in the sign convention and $-\sigma_t$ is equivalent to the mean total compressive stress P used in the phenomenological approach.[27,33] Taking time derivatives on both sides of eq 9 and combining with eq 8 to eliminate dp_s/dt yields

$$\left(1+\frac{p-p_s}{K_s}\right)\frac{d\phi}{dt} = (1-\phi)\nabla\cdot\boldsymbol{v}_s - \frac{1}{K_s}\left(\frac{d\sigma_t}{dt} + \phi\frac{dp}{dt}\right) + \frac{1-\phi}{K_x}\frac{dp}{dt} \tag{10}$$

Until now, the effects of anisotropy and micro-inhomogeneity have not been addressed, which are discussed below during the derivation of the effective stress equation.

Consider the general loading case for a REV of a fluid-saturated porous medium; i.e., the sample is subjected to a total stress $\boldsymbol{\sigma}$, and the interconnected pores are under the fluid pressure p. We make use of the principle of superposition to decompose this general scenario into a drained test (I) and an unjacketed test (II), which are described as follows.

- For the drained test (I), the fluid pressure is zero $p^{(I)} = 0$, while the sample is subjected to the stress $\boldsymbol{\sigma}^{(I)} = \boldsymbol{\sigma} + p\mathbf{1}$. Note that the boldfaced $\mathbf{1}$ represents the second-order identity tensor. By using the fourth-order drained bulk stiffness tensor $\mathbb{M}$, the corresponding strain $\boldsymbol{\varepsilon}^{(I)}$ could be calculated as

$$\boldsymbol{\varepsilon}^{(I)} = \mathbb{M}^{-1}:\boldsymbol{\sigma}^{(I)} = \mathbb{M}^{-1}:(\boldsymbol{\sigma} + p\mathbf{1}) \tag{11}$$

  The elastic anisotropy is now embedded into $\mathbb{M}$. In this work, we consider a special case of anisotropy named transverse isotropy, and $\mathbb{M}$ is controlled by five elastic constants and the vector $\boldsymbol{n}$ normal to the plane of isotropy. In geology, the plane of isotropy is also known as the bedding plane.[31] For detailed expressions of $\mathbb{M}$ and $\mathbb{M}^{-1}$, please refer to Section 4.1 in Zhang and Borja[32] or an in-depth tutorial by Nemeth.[37]
- For the unjacketed test (II), we assume the sample is submersed in a fluid chamber, so both the exterior solid surface and the interior surface opening to the pores are exposed to the same normal stress, *i.e.*, $p^{(II)} = p$ and $\boldsymbol{\sigma}^{(II)} = -p\mathbf{1}$. For this scenario, if the solid is homogeneous (made by one kind of mineral) and without occluded pores, the sample deforms in a self-similar fashion that leads to unchanged porosity. However, as pointed out by Brown and Korringa,[28] the solid of sedimentary rocks could contain more than one mineral, particularly an amount of clay, which is much softer. In addition, the occluded pores and fissures are common. In order to describe this disproportionate change due to micro-inhomogeneity and recall the anisotropic nature, we propose a fourth-order unjacketed bulk compliance tensor $\mathbb{C}_s'$, and the corresponding strain $\boldsymbol{\varepsilon}^{(II)}$ is

$$\boldsymbol{\varepsilon}^{(II)} = \mathbb{C}_s':\boldsymbol{\sigma}^{(II)} = -(\mathbb{C}_s':\mathbf{1})p \tag{12}$$

  The physical significance of $\mathbb{C}_s'$ and its relation with K_s would become more clear in the next section.

From the above analysis, the summation of $\boldsymbol{\varepsilon}^{(I)}$ and $\boldsymbol{\varepsilon}^{(II)}$ represents the actual strain tensor $\boldsymbol{\varepsilon}$

$$\boldsymbol{\varepsilon} = \mathbb{M}^{-1}:(\boldsymbol{\sigma} + p\mathbf{1}) - (\mathbb{C}_s':\mathbf{1})p \tag{13}$$

Left-multiplying both sides by $\mathbb{M}$ yields the effective stress tensor $\boldsymbol{\sigma}''$

$$\boldsymbol{\sigma}'' = \mathbb{M}:\boldsymbol{\varepsilon} = \boldsymbol{\sigma} + p\mathbf{1} - \mathbb{M}:(\mathbb{C}_s':\mathbf{1})p = \boldsymbol{\sigma} + (\mathbf{1} - \mathbb{M}:\mathbb{C}_s':\mathbf{1})p \tag{14}$$

where we identify the Biot tensor (an upgraded version of the Biot coefficient scalar[26]) as

$$\boldsymbol{\alpha} \overset{\text{def}}{=} \mathbf{1} - \mathbb{M}:\mathbb{C}_s':\mathbf{1} \tag{15}$$

Now, if we take the trace of eq 14, followed by the time derivative, we would obtain

$$\frac{d\sigma_t}{dt} = \frac{\mathbf{1}:\mathbb{M}}{3}:\frac{d\boldsymbol{\varepsilon}}{dt} - \frac{\mathbf{1}:\boldsymbol{\alpha}}{3}\frac{dp}{dt} \tag{16}$$

which is then substituted into eq 10 to get

$$\left(1+\frac{p-p_s}{K_s}\right)\frac{d\phi}{dt} + \phi\nabla\cdot\boldsymbol{v}_s = \left(\mathbf{1} - \frac{\mathbf{1}:\mathbb{M}}{3K_s}\right):\frac{d\boldsymbol{\varepsilon}}{dt} + \left[\frac{1}{K_s}\left(\frac{\mathbf{1}:\boldsymbol{\alpha}}{3} - \phi\right) + \frac{1-\phi}{K_x}\right]\frac{dp}{dt} \tag{17}$$

According to the expression of $\boldsymbol{\alpha}$ in eq 15, eq 17 could also be reformulated as

$$\left(1+\frac{p-p_s}{K_s}\right)\frac{d\phi}{dt} + \phi\nabla\cdot\boldsymbol{v}_s = \left(\mathbf{1} - \frac{\mathbf{1}:\mathbb{M}}{3K_s}\right):\frac{d\boldsymbol{\varepsilon}}{dt} + \left[\frac{1}{K_s}\left(1 - \frac{\mathbf{1}:\mathbb{M}:\mathbb{C}_s':\mathbf{1}}{3} - \phi\right) + \frac{1-\phi}{K_x}\right]\frac{dp}{dt} \tag{18}$$

Equation 18 is the finalized form for the mass balance of solid, which is rigorously derived in a manner of mixture theory without any approximations. However, there is a factor $(p-p_s)/K_s$ in front of $d\phi/dt$, which prevents further simplification. Therefore, we make the following two assumptions regarding this term and infinitesimal deformation:

1 The order of $(p-p_s)/K_s$ is much less than 1. Thus, the left-hand side of eq 18 is simply $d\phi/dt + \phi\nabla\cdot\boldsymbol{v}_s$.
2 In the infinitesimal deformation, the deformation Jacobian J is very close to 1. Thus, the time derivative of Lagrangian porosity $\phi^L = J\phi$ can be approximated by $d\phi/dt + \phi\nabla\cdot\boldsymbol{v}_s$ (recall that $dJ/dt = J\nabla\cdot\boldsymbol{v}_s$). As a result, the left-hand side of eq 18 is indeed $d\phi^L/dt$.

It is worth mentioning that the above analysis is not dependent on the specific fluid type. In fact, in addition to the liquid, inert

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gas such as helium or nitrogen has been used to measure pore volume change under changing effective stress condition.[38,39] Therefore, the following numerical analysis for shale gas indeed makes sense.

Strong form. Before closing this section, it would be beneficial to summarize the complete hydro-mechanical governing equations of shale gas flow (gravity is ignored). We consider a closed domain $\overline{\mathcal{B}}=\mathcal{B}\cup\partial\mathcal{B}$, where $\mathcal{B}$ is an open domain and $\partial\mathcal{B}$ is the boundary. As usual, the boundary could be decomposed into essential and natural parts $\partial\mathcal{B}_u$ and $\partial\mathcal{B}_t$, where displacement $\boldsymbol{u}$ and traction $\boldsymbol{t}$ are prescribed, respectively; and $\partial\mathcal{B}_p$ and $\partial\mathcal{B}_q$, where gas pressure p and flux q are prescribed, respectively. The following relations hold ($\emptyset$ implies the empty set)

$$\overline{\partial\mathcal{B}_u\cup\partial\mathcal{B}_t} = \overline{\partial\mathcal{B}_p\cup\partial\mathcal{B}_q}=\partial\mathcal{B} \tag{19}$$

$$\partial\mathcal{B}_u\cap\partial\mathcal{B}_t=\partial\mathcal{B}_p\cap\partial\mathcal{B}_q = \emptyset \tag{20}$$

The strong form of the initial boundary-value problem is as follows. Find $\boldsymbol{u}$: $\overline{\mathcal{B}}\times[0, +\infty)\to\mathbb{R}^{\mathrm{sd}}$ (sd = 2 or 3 is the space dimension) and p: $\overline{\mathcal{B}}\times[0, +\infty)\to\mathbb{R}$ such that "in $\mathcal{B}$"

$$\nabla\cdot(\boldsymbol{\sigma}'' - \boldsymbol{\alpha}p) + \boldsymbol{b} = \boldsymbol{0} \tag{21}$$

$$\frac{\partial(\rho_{\mathrm{g}}\phi^{L})}{\partial t}+\nabla\cdot\underbrace{\left(-\frac{\rho_{\mathrm{g}}\boldsymbol{k}}{\mu_{\mathrm{g}}}\cdot\nabla p\right)}_{\boldsymbol{F}=\rho_{\mathrm{g}}\boldsymbol{q}} = -(1-\phi_0)\frac{\partial m}{\partial t} \tag{22}$$

where $\boldsymbol{b}$ is the prescribed body force vector ($\boldsymbol{0}$ by default), $\boldsymbol{F}$ is known as the mass flux vector, $\boldsymbol{q}$ is the Darcy flux vector, $\boldsymbol{k}$ is the permeability tensor, μ_{g} is the gas viscosity, ϕ_0 is the initial porosity, ρ_{g} is the gas density, and m is the mass of adsorbed gas per solid volume.[40] Both ρ_{g} and m are functions of gas pressure p as

$$\rho_{\mathrm{g}} = \frac{pM_{\mathrm{g}}}{ZRT}, \qquad m = \rho_{\mathrm{r}}\rho_{\mathrm{gstd}}V_{\mathrm{L}}\frac{p/Z}{P_{\mathrm{L}}+p/Z} \tag{23}$$

In eq 23, M_{g} is the gas molar mass, Z is the gas compression factor,[41] R is the universal gas constant, T is the reservoir temperature, ρ_r is the constant rock bulk density, ρ_{gstd} is the gas density at the standard condition, and V_{L} and P_{L} are the Langmuir volume and Langmuir pressure, respectively.

Finally, the strong form also requires the satisfaction of the boundary conditions on $\partial\mathcal{B}$ where $\boldsymbol{n}_\partial$ represents the outward normal vector

$$\boldsymbol{u} = \hat{\boldsymbol{u}} \quad \text{on} \quad \partial\mathcal{B}_u \tag{24}$$

$$\boldsymbol{n}_\partial\cdot\boldsymbol{\sigma} = \hat{\boldsymbol{t}} \quad \text{on} \quad \partial\mathcal{B}_t \tag{25}$$

$$p = \hat{p} \quad \text{on} \quad \partial\mathcal{B}_p \tag{26}$$

$$-\boldsymbol{q}\cdot\boldsymbol{n}_\partial = \hat{q} \quad \text{on} \quad \partial\mathcal{B}_q \tag{27}$$

and initial conditions

$$\boldsymbol{u}(\boldsymbol{x}, t=0) = \boldsymbol{u}_0(\boldsymbol{x}), \qquad p(\boldsymbol{x}, t=0) = p_0(\boldsymbol{x}) \tag{28}$$

for all $\boldsymbol{x}\in\overline{\mathcal{B}}$.

3. BOUNDS OF MATERIAL CONSTANTS AND THEIR PHYSICAL SIGNIFICANCE

3.1. General Case. In this section, we explore the bounds of various material constants and their combinations that yield other physical measurable constants from experiments. First of all, based on the thermodynamics consideration of particle interactions,[33] the effective bulk modulus $\overline{K}_{\mathrm{s}}$ of the composite solid, which includes more than one mineral and entrapped fluid in occluded pores, must be positive. Hence, we establish that

$$\overline{K}_{\mathrm{s}}>0 \tag{29}$$

Since the solid is made by more than one mineral, the Voigt bound gives an upper bound of $\overline{K}_{\mathrm{s}}$, which is the weighted sum of each mineral's bulk modulus $K_{\mathrm{m},i}$ (the weight is each mineral's volume fraction v_i).[33] If there is a dominant mineral with bulk modulus $\overline{K}_{\mathrm{m}}$, then, we could get a looser bound:

$$\overline{K}_{\mathrm{s}} \le \sum_i v_i K_{\mathrm{m},i} \le \sum_i v_i\overline{K}_{\mathrm{m}} = \overline{K}_{\mathrm{m}} \tag{30}$$

Second, as for the elastic stability condition, if we recall the positiveness of the elastic strain energy, we should establish the requirement that $\mathbb{M}$ (with major and minor symmetry) should be positive-definite. According to the definition given in Song,[42] a fourth-order symmetric tensor M_{ijkl} is said to be positive-definite if and only if for any nonzero symmetric second-order tensor ξ_{ij}

$$M_{ijkl}\xi_{ij}\xi_{kl}>0 \tag{31}$$

Similarly, the elastic energy could also be written in terms of the compliance tensor $\mathbb{M}^{-1}$, so we can deduce that $\mathbb{M}^{-1}$ is also positive-definite.

Third, from the strain decomposition in eq 13, the volumetric strain $\epsilon_v = \Delta V/V$ can be calculated as

$$\frac{\Delta V}{V} = \boldsymbol{1}:\mathbb{M}^{-1}:\boldsymbol{\sigma} + [\boldsymbol{1}:(\mathbb{M}^{-1} - \mathfrak{c}_{\mathrm{s}}'):\boldsymbol{1}]p \tag{32}$$

If we further assume $\boldsymbol{\sigma} = -P\boldsymbol{1}$ where P is the isotropic total compressive load, then, we recover a similar form as that in the standard poroelasticity[43,44]

$$\frac{\Delta V}{V} = -(\boldsymbol{1}:\mathbb{M}^{-1}:\boldsymbol{1})P' - (\boldsymbol{1}:\mathfrak{c}_{\mathrm{s}}':\boldsymbol{1})p = -C^*P' - C_{\mathrm{s}}'^*p \tag{33}$$

where P' is defined as $P' = P - p$, $C^* = \boldsymbol{1}:\mathbb{M}^{-1}:\boldsymbol{1}$ is known as the generalized drained compressibility,[30] and $C_{\mathrm{s}}'^* = \boldsymbol{1}:\mathfrak{c}_{\mathrm{s}}':\boldsymbol{1}$ is known as the generalized unjacketed bulk compressibility.[30] Since $\mathbb{M}^{-1}$ is positive-definite, we know $C^*>0$. Furthermore, the value of $C_{\mathrm{s}}'^*$ has been routinely measured for various rocks in the past several decades,[45,46] and there was no indication that this value could be negative. Therefore, we additionally establish that

$$C_{\mathrm{s}}'^*>0 \tag{34}$$

Finally, for practical materials, we assume that the drained porosity compressibility C_ϕ is positive.[33] In order to derive an expression for C_ϕ, we start from eq 10 and set $p\equiv 0$ and $\sigma_t\equiv -P$; we have

$$\frac{\mathrm{d}\phi}{\mathrm{d}t} = (1-\phi)\nabla\cdot\boldsymbol{v}_{\mathrm{s}} + \frac{1}{K_{\mathrm{s}}}\frac{\mathrm{d}P}{\mathrm{d}t} \tag{35}$$

As $\nabla\cdot\boldsymbol{v}_{\mathrm{s}}$ represents the time derivative of volumetric strain $\Delta V/V$, by using eq 33 and the fact that $p\equiv 0$, we have

$$\nabla\cdot\boldsymbol{v}_{\mathrm{s}} = -C^*\frac{\mathrm{d}P}{\mathrm{d}t} \tag{36}$$

and

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$$\frac{\mathrm{d}\phi}{\mathrm{d}t} = -\left[(1-\phi)C^* - \frac{1}{K_s}\right]\frac{\mathrm{d}P}{\mathrm{d}t} \tag{37}$$

An implicit backward integration of eq 37 yields (P starts from 0)

$$\frac{\Delta\phi}{1-\phi} = -\underbrace{\left[C^* - \frac{C_s}{1-\phi}\right]}_{C_\phi} P \tag{38}$$

where $C_s = 1/K_s$. Based on the condition that $C_\phi > 0$, we obtain the constraint

$$C^* > \frac{C_s}{1-\phi} \tag{39}$$

In Cheng,[33] the author defined a new intrinsic poroelasticity constant C_ψ known as the unjacketed porosity compressibility. Therefore, it will be interesting to see how to represent C_ψ using poroelastic constants in this work under anisotropy. We consider the scenario of an unjacketed test where $p = p_s = -\sigma_t$ (this is consistent with the original definition of $P' = 0$); after incorporating this condition, eq 10 becomes

$$\begin{aligned}\frac{\mathrm{d}\phi}{\mathrm{d}t} &= (1-\phi)\nabla\cdot\boldsymbol{v}_s + \left(\frac{1-\phi}{K_s} + \frac{1-\phi}{K_x}\right)\frac{\mathrm{d}p}{\mathrm{d}t}\\ &= (1-\phi)\nabla\cdot\boldsymbol{v}_s + \frac{1-\phi}{\bar{K}_s}\frac{\mathrm{d}p}{\mathrm{d}t}\end{aligned} \tag{40}$$

In order to replace $\nabla\cdot\boldsymbol{v}_s$, we make use of eq 33 again. Since $P' = 0$, we have

$$\nabla\cdot\boldsymbol{v}_s = -C_s'^*\frac{\mathrm{d}p}{\mathrm{d}t} \tag{41}$$

and

$$\frac{\mathrm{d}\phi}{\mathrm{d}t} = (1-\phi)\left(\frac{1}{\bar{K}_s} - C_s'^*\right)\frac{\mathrm{d}p}{\mathrm{d}t} \tag{42}$$

An implicit backward integration of eq 42 yields (p starts from 0)

$$\frac{\Delta\phi}{1-\phi} = -\left(C_s'^* - \frac{1}{\bar{K}_s}\right)p \tag{43}$$

from which we can identify the generalized C_ψ as

$$C_\psi = C_s'^* - \frac{1}{\bar{K}_s} \tag{44}$$

At this point, we can give a summary of the above findings

- $\bar{K}_s$ satisfies the constraint $0 < \bar{K}_s \le \bar{K}_m$.
- Fourth-order tensors $\mathbb{M}$ and $\mathbb{M}^{-1}$ are positive-definite.
- The compressibilities have the following inequalities:

$$C_s'^* > 0 \quad \text{and} \quad C^* > C_s/(1-\phi)$$

- C_ψ can be expressed through eq 44.
- Suppose the frame deformation of porous material can be measured the same way as that of an anisotropic elastic solid, we could obtain $\mathbb{M}$ and $\mathfrak{c}_s':\mathbf{1}$ through drained and unjacketed tests, respectively, which subsequently lead to the determination of $C_s'^*$, C^*, and $\boldsymbol{\alpha}$ through simple tensor calculation.
- If the measurement of $\Delta\phi$ could also be obtained through drained and unjacketed tests (which could be difficult, especially for the unjacketed test), then from the value of C_ψ and C_ϕ, K_s and $\bar{K}_s$ could be obtained, which also determines K_x through eq 7. Therefore, theoretically speaking, we could obtain the complete set of anisotropic poroelastic constants that appear in eq 18 through experiments.

3.2. Specific Case of Isotropy. In this section, we carry the analysis further by assuming isotropy, a widely used scenario in literature. As a result, we have

$$\mathfrak{c}_s':\mathbf{1} = \frac{\mathbf{1}}{3K_s'}, \qquad C_s'^*:\mathbf{1}:\mathfrak{c}_s':\mathbf{1} = \frac{1}{K_s'} = C_s' \tag{45}$$

where K_s' is the unjacketed bulk modulus,[44] and in isotropy we could drop the "*" superscript sign in several quantities such as $C_s'^*$. The Biot tensor would also downgrade to a scalar since

$$\mathbb{M}:\mathfrak{c}_s':\mathbf{1} = \frac{\mathbb{M}:\mathbf{1}}{3K_s'} = \frac{K}{K_s'}\mathbf{1} \tag{46}$$

where K is the drained bulk modulus and it is indeed $1/C^*$ in isotropy. Hence, the Biot coefficient scalar α is given as[44]

$$\alpha = 1 - \frac{K}{K_s'} \tag{47}$$

Through this part, we aim to find the relation among K_s, K_x, K_s', and K_s'', in which K_s'' is known as the unjacketed pore modulus in the literature.[33] To achieve that, we make use of the expression of the so-called Biot modulus. On the one hand, according to eq 18, the Biot modulus N is given as (note that we exclude the fluid compressibility term ϕ/K_f)

$$\begin{aligned}\frac{1}{N} &= \frac{1}{K_s}\left(1 - \frac{\mathbf{1}:\mathbb{M}:\mathfrak{c}_s':\mathbf{1}}{3} - \phi\right) + \frac{1-\phi}{K_x}\\ &= \frac{\alpha-\phi}{K_s} + \frac{1-\phi}{K_x}\end{aligned} \tag{48}$$

On the other hand, the standard poroelasticity micromechanical analysis gives the following result, as shown in Table 1 of Detournay and Cheng[44]

$$\frac{1}{N} = \frac{K}{K_s'}\left(\frac{1}{K} - \frac{1}{K_s'}\right) - \frac{\phi}{K_s''} = \frac{\alpha}{K_s'} - \frac{\phi}{K_s''} \tag{49}$$

By simultaneously considering the above two equations, we obtain the relation

$$\frac{\alpha-\phi}{K_s} + \frac{1-\phi}{K_x} = \frac{\alpha}{K_s'} - \frac{\phi}{K_s''} \tag{50}$$

To solve for K_s and K_x, we definitely need another equation. On the one hand, by matching C_ϕ with eq 27 in Detournay and Cheng,[44] we get $K_s = K_s'$. Thus, K_x can be solved from eq 50, which gives

$$\frac{1}{K_x} = \frac{\phi}{1-\phi}\left(\frac{1}{K_s'} - \frac{1}{K_s''}\right) \tag{51}$$

By using eq 7 and eq 44, we could get

$$\bar{C}_s = \frac{C_s' - \phi C_s''}{1-\phi} \text{ and } C_\psi = \frac{1}{K_\psi} = \frac{\phi(C_s'' - C_s')}{1-\phi} \tag{52}$$

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Table 1. Comparisons of $\overline{K}_s$ and K_ψ on Surviving Cases Reported in Cheng[33] by Using Different Formulas[a]

Rock type[ref]	P' (MPa)	ϕ	K_s' (GPa)	K_s'' (GPa)	K (GPa)	α	$\overline{K}_m$ (GPa)	From eq 52		From eq 53	
								$\overline{K}_s$ (GPa)	K_ψ (GPa)	$\overline{K}_s$ (GPa)	K_ψ (GPa)
Lavoux limestone[47]	2	0.23	44	130	8.3	0.811	77	36.74	−222.67	34.87	−168.12
Vilhanneur limestone[47]	2	0.14	56	110	2	0.964	77	51.86	−700.74	51.69	−671.64
Bentheimer sandstone[48]	5−30	0.24	37	59	14	0.622	37	33.10	−314.22	29.97	−157.78
Berea sandstone[49]	5−20	0.23	29	22	14	0.517	37	32.05	305.13	38.91	113.83
Opalinus clay (shale)[36]	0.5	0.13	8.9	7	1.85	0.792	—	9.28	219.44	9.40	167.01

[a]"—": No data.

where $\overline{C}_s=1/\overline{K}_s$, $C_s' = 1/K_s'$, and $C_s'' = 1/K_s''$. This result is identical to that in Cheng.[33] Note that Cheng[33] denoted $\overline{K}_s$ as K_s in his paper. On the other hand, we could get an approximate solution by letting $K_x = \infty$, which implies $\overline{K}_s = K_s$ and eq 6 recovers the commonly used barotropic law.[31,34] As a result, we get the following expressions

$$\overline{C}_s = \frac{\alpha C_s' - \phi C_s''}{\alpha - \phi} \text{ and } C_\psi = \frac{1}{K_\psi} = \frac{\phi(C_s'' - C_s')}{\alpha - \phi} \tag{53}$$

Now, it would be beneficial to compare eq 52 with eq 53. Differences could be observed since α in eq 53 has been replaced by 1 in eq 52. Nevertheless, for many rock materials, both formulas will generate close results, as shown in Table 1 applied to the surviving cases in Cheng,[33] and for an ideal porous medium in which $C_s'' = C_s'$, the results are exactly the same. Although Cheng[33] first discovered these intrinsic constants, our work makes the first attempt to recover composite solid modulus $\overline{K}_s$ through mixture theory.

Remark. Although in Sections 2 and 3, we assume the most general condition, in the following numerical simulations, we still adopt the ideal porous medium assumption due to lack of reasonable complete input data, so $K_x = K_\psi = \infty$, and there will be no distinction among K_s, $\overline{K}_s$, K_s', and K_s''. Hence, we simply denote them as K_s (solid grain bulk modulus[25]) for brevity.

4. MODEL VERIFICATION OF 3D ELASTIC ANISOTROPY

As the novelty of this work lies in the incorporation of elastic anisotropy, we have designed this verification example to test the implementation of anisotropy in a 3D hydro-mechanical coupling finite element framework. The computational domain is $[0, 1] \times [0, 1] \times [0, 1]$ m^3, as shown in Figure 1, together with the finite element mesh. The problem is designed to be a quasi-static problem. The material properties and boundary conditions are detailed as follows.

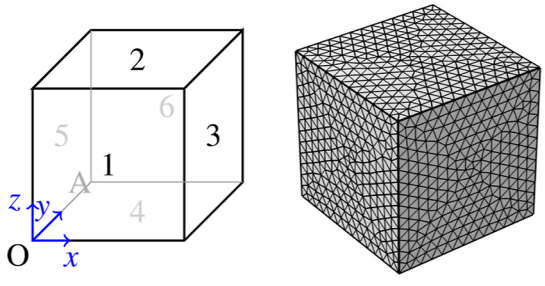


Figure 1. Schematic of the computational domain with all the boundaries labeled from 1 to 6 and the finite element mesh is shown (tetrahedron element) on the right.

- For the solid part, we restrict the normal displacement on the boundary surface 4, i.e., a roller boundary condition. In order to eliminate any rigid body motion, we fix the node at the origin (Point O in Figure 1) and prescribe $u_x = 0$ at Point A in Figure 1. For the boundary surfaces 1, 3, 5, and 6, the normal component of the traction is $\sigma_n = -\sqrt{4 - 3z}$ MPa (in compression) and the shear component is $t_n = 0$. For the top boundary surface 2, the normal component of the traction is $\sigma_n = -3$ MPa (in compression) and the shear component is $t_n = 0$. We also have a non-trivial body force field (in MPa/m)

$$\boldsymbol{b} = -\left\{0 \quad 0 \quad \frac{3}{2\sqrt{4 - 3z}}\right\}^T \tag{54}$$

  For the solid parameters, we set $E_v = 5$ MPa, $E_h = 10$ MPa, $\nu_{hh} = 0.2$, $\nu_{vh} = 0.25$, $G_{vh} = 4$ MPa, $n_x = -0.5$, $n_y = 0$, $n_z = \sqrt{3}/2$, and $K_s = 75$ GPa. Note that K_s satisfies eq 30 if the dominant mineral is calcite ($\overline{K}_m$=77 GPa).
- For the gas flow part, the inlet pressure is prescribed on the boundary surface 4 with a value of 2 MPa, and the outlet pressure is prescribed on the boundary surface 2 with a value of 1 MPa. The other four surfaces have zero flux. For the gas flow parameters, as long as the gas belongs to the type of ideal gas, and permeability, viscosity, and gas molar mass are constant, the results are the same.

Under the above settings, the analytical solution for gas pressure would be (in MPa)

$$p = \sqrt{4 - 3z} \tag{55}$$

The displacement field would be identical to a **drained** scenario when we apply a compressive load of 2 MPa on the boundary surface 2 of this transversely isotropic material with zero lateral stress and roller support for the bottom surface. The verification results are shown in Figure 2 for both displacement and gas pressure, and a perfect match can be observed, which validates the correct implementation of elastic anisotropy. An evident feature of elastic anisotropy is that even when the "effective" mechanical loading is in the z direction, it will swing in the x direction, see Figure 2a. This result is consistent with Figures 9A and 10A in Zhao et al.[50]

5. MODEL APPLICATION

5.1. Model Setup. For the model application, a three-dimensional anisotropic model is established to analyze the effects of anisotropy, as shown in Figure 3. A hydraulic fracture filled with proppant[51] is located at the center of the computational domain. The proppant deformation follows the elastoplastic Drucker-Prager model.[52,53] A production well is

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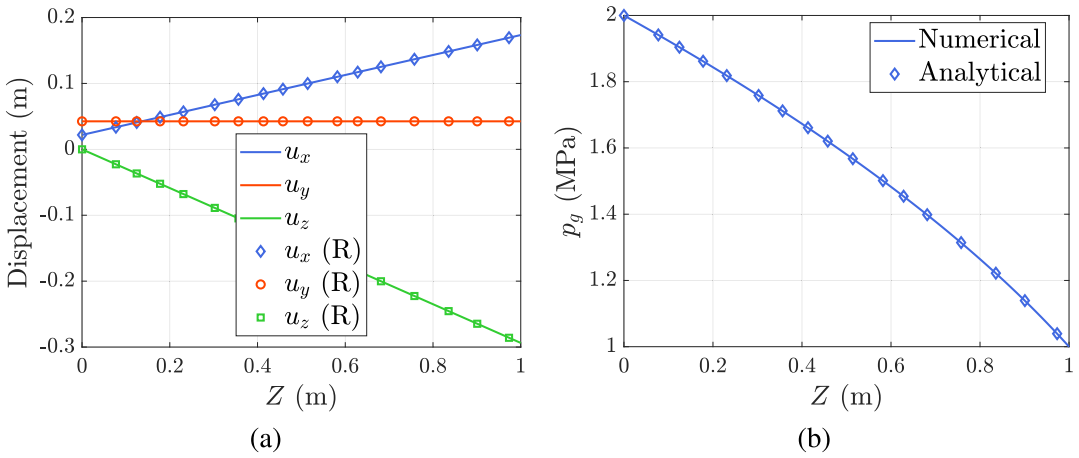


Figure 2. (a) Verification of the displacement field along the center vertical line. "(R)" represents the reference solution in a drained scenario, i.e., pure solid deformation. (b) Verification of the gas pressure along the same line.

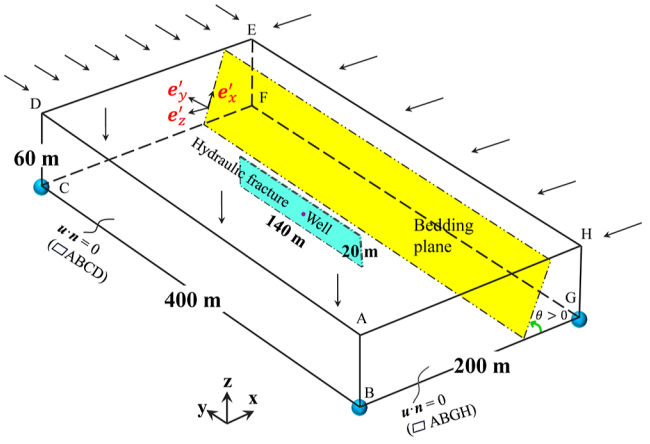


Figure 3. Schematic of an anisotropic 3D shale gas reservoir. The yellow plane represents the bedding.

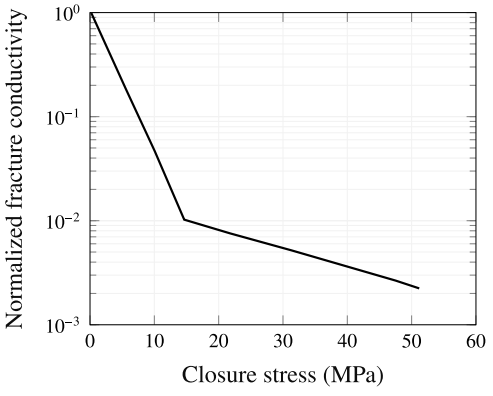


Figure 4. Normalized fracture conductivity under different "effective" closure stresses. The significant fracture conductivity loss could be explained from the perspective of formation damage[55] induced by proppant embedment, residual fracturing fluid, or proppant fines migration.[55]

located at the center of the hydraulic fracture. For the boundary conditions, we assume zero flux across all of the external faces, and the constant compressive load 20 MPa is applied on the top (ADEH), right (EFGH), and back (CDEF) faces. The remaining three faces have a roller boundary condition that restricts normal displacement. The stress-dependent normalized fracture conductivity is shown in Figure 4, and the other parameters are listed in Table 2. In our modeling process, the hydraulic fracture is treated as a 3D object whose thickness is equal to the initial fracture aperture w_{F0}. As a result, the strong form of Section 2 still applies. Note that since hydraulic fracture has a much larger permeability and a much smaller storage capacity than the surrounding matrix, we could drop the time derivative terms in eq 22. For the well, it is treated as a 2D object with a dimension $2r_w \times w_{F0}$ where r_w is the well radius and w_{F0} is the initial fracture aperture, and it also acts as a Dirichlet boundary with prescribed bottom-hole pressure (BHP). Finally, to verify the correctness of the simulation results, mesh sensitivity analysis is also performed. The mesh is shown in Figure 5. We adopt the quadratic serendipity element for the displacement field and linear element for the gas pressure field, which satisfy the inf-sup stability condition.[54]

Since this is definitely a transient problem, the software has adopted an adaptive time-stepping algorithm with an initial step around 0.14 s and the time amplification factor $\Delta t_{n+1}/\Delta t_n = 1$ or 2. The total number of steps taken by the segregated solver is around 60. The data stored in output are interpolated between the time steps taken by the solver.

5.2. Model Results. The mesh sensitivity analysis result is depicted in Figure 6. Here, we choose the quantity of residual fracture aperture to verify our result, which should not change under further mesh refinement, and Figure 6 is consistent with our expectation. Another feature that we could observe is that our model captures the displacement discontinuity across the fracture, which is not considered in conventional shale gas reservoir modeling. As verified by Yan et al.,[58] the neglect of displacement discontinuity would lead to an underestimate of the cumulative gas production. We note that this residual fracture aperture shape may not align with reality. This inconsistency arises from our computational model in which we assume an initial uniform aperture. Due to the strong end constraints from the rock matrix, the aperture at the fracture tips does not decrease, creating tiny stress singularity points near the fracture tip. Nevertheless, these tiny stress singularity points do not affect the overall stress distribution, and many published studies have the same treatment.[59−61]

We next discuss the impact of elastic anisotropy on the stress distribution, proppant deformation, and cumulative gas production. To make the discussion meaningful, we need to

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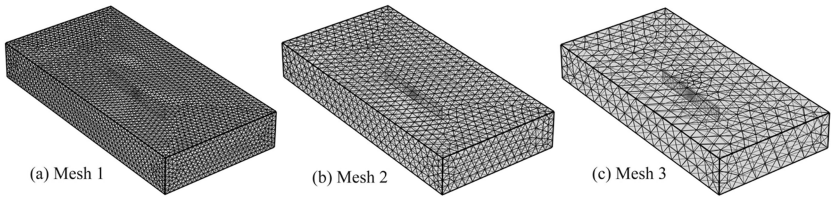


Figure 5. Three different mesh sizes in the mesh sensitivity analysis of the 3D problem. In the domain of fracture, we adopt the prism element, while for the remaining domain, the tetrahedron element is adopted. By default, mesh 2 is used in the remaining analysis.

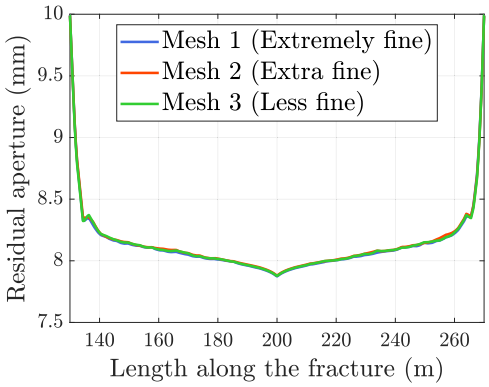


Figure 6. Mesh sensitivity analysis result of the residual fracture aperture at z = 30 m and t = 1000 days. The parameters are based on Table 2 with $\theta = 0$.

compare it with the case of elastic isotropy. Thus, we design another set of parameters from Table 2 as

- E = 40 GPa and ν = 0.25. K_s remains the same. Note that the isotropic elastic parameters cannot be chosen arbitrarily; they should be as close as possible to the anisotropic parameters in some ways. Here, we remark that the bulk modulus $K = E/3/(1-2\nu)$ is close to the generalized bulk modulus of the anisotropic elasticity $K^* = (\mathbf{1}:\mathbb{M}:\mathbf{1})/9$;[30] and the shear modulus $G = E/2/(1+\nu)$ is close to G_{vh}. In addition, the value of K_s satisfies eq 30 if the dominant mineral is pyrite ($\bar{K}_m$=147 GPa).

Figure 7 shows the total stress σ_{xx} distribution at one and half years after production, and Figure 8 shows the temporal response of σ_{xx} at two representative locations. Both figures suggest that elastic anisotropy plays a significant role, and the simplified model based on the constant total stress assumption may lead to errors.

First of all, for the middle layer, the isotropic case has the largest decrease in $|\sigma_{xx}|$ at the central region surrounding the fracture. Second, for the top layer, the increase in $|\sigma_{xx}|$ at the central region for the three "TI" cases could be either larger or smaller than that of the isotropic case, depending on the bedding plane orientation θ. Third, when $\theta = \pi/4$, the boundary effect on $|\sigma_{xx}|$ is significant on the top layer, and this explains why the green curve in Figure 8a has a more rapid decrease than the yellow curve at the end of the simulation.

The above findings certainly remind us of the "stress shadow" effect.[62−64] As mentioned in the literature, the creation of a fracture generates a perturbation in the surrounding stress regime, affecting the propagation of other fractures.[64] This interference phenomenon is known as the "stress shadow" effect. During the simultaneous multiple fracture initiation, the combined stress shadow effects of the propagating fractures make them grow differently.[63] Although Figure 7 is for the production stage, we deduce that the elastic anisotropy also affects the stress in the fracture initiation and propagation stage. As a result, the interplay between the stress shadow and elastic anisotropy could make the prediction of fracture propagation even more difficult. Also, the boundary effect on stress distribution is much more severe for $\theta \approx \pi/4$, which should also be taken into consideration. The practical implications can be summarized as follows. A field investigation of the degree of anisotropy is important for the design of hydraulic fracturing, and we should have countermeasures when the propagation of hydraulic fracture deviates significantly from the design due to anisotropy.

While the bedding plane orientation has an evident effect on the horizontal stress parallel to the wellbore (in our case, σ_{xx}), its effect on σ_{yy} is less significant, as shown by Figure 9. This finding is quite consistent with the findings from the work on rock anisotropy,[4,10,17,25] i.e., as opposed to the habitually applied isotropic assumption, the horizontal stress parallel to the wellbore shows a strong dependency on the anisotropic

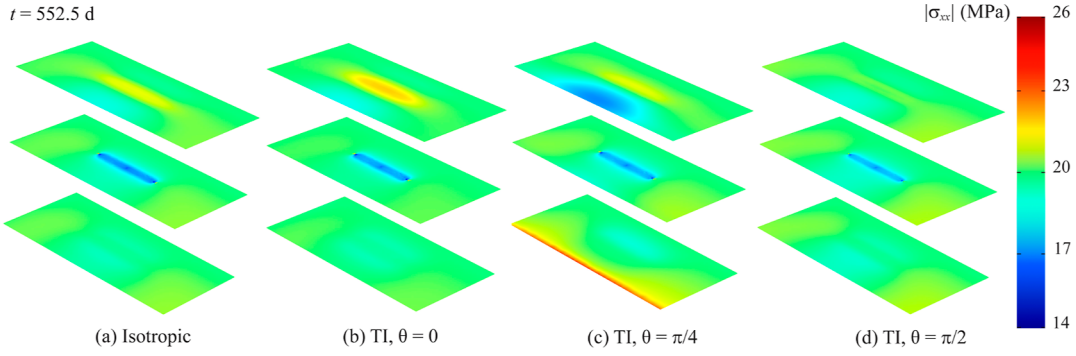


Figure 7. Comparisons of the total stress σ_{xx} field for isotropic and transversely isotropic shale. The profiles are shown for the top (z = 60 m), middle (z = 30 m), and bottom layers (z = 0 m). "TI" is abbreviated for "Transversely Isotropic".

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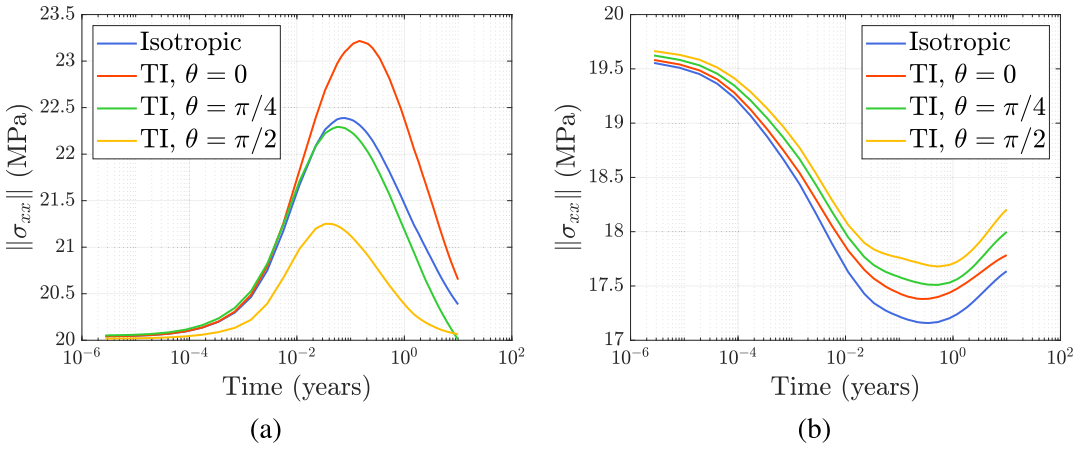


Figure 8. (a) Total stress σ_{xx} evolution at the observation point (100, 215, 60) m. (b) Total stress σ_{xx} evolution at the observation point (100, 215, 30) m.

Table 2. Simulation Parameters for the 3D Shale Reservoir with a Propped Hydraulic Fracture

Parameter	Value	Unit
Initial absolute permeability of matrix k_0	10^{-17}	m^2
Initial absolute permeability of hydraulic fracture k_{F0}	10^{-11}	m^2
Initial aperture of hydraulic fracture w_{F0}	0.01	m
Initial porosity of matrix and hydraulic fracture	0.05, 0.5	1
Langmuir pressure P_L	4	MPa
Langmuir volume V_L	0.018	m^3/kg
Initial reservoir pressure and BHP	25, 10	MPa
Gas viscosity μ_g	Ref 56	Pa·s
Reservoir temperature T	343.15	K
Rock density ρ_r	2850	kg/m^3
Well radius r_w	0.1	m
Compressive load (traction)	20	MPa
Proppant mechanical properties: E_{prop} = 40 MPa, ν_{prop} = 0.001, c_{prop} = 1 MPa, $\phi_{prop} = \psi_{prop}$ = 40°, α = 0.8		N.A.
Shale matrix mechanical properties:[16,57] E_v = 20 GPa, E_h = 40 GPa, ν_{hh} = 0.2, ν_{vh} = 0.25, G_{vh} = 15 GPa, K_s = 137.3735 GPa		N.A.

poroelastic properties, and this stress would control hydraulic fracture geometries and proppant concentration.[4,17]

The elastic anisotropy also affects the proppant deformation and cumulative gas production, as revealed by Figures 10 and 11. Due to the largest decrease of $|\sigma_{xx}|$ in the isotropic case, it has the least fracture compression and plastic strain. Also, since small $|\sigma_{xx}|$ leads to small "effective" closure stress, according to the fracture conductivity law in Figure 4, the conductivity loss is also small, which explains the highest gas cumulative production in Figure 11. A similar argument is also applicable to the cases of vertical and horizontal bedding. Now, suppose we ignore the displacement discontinuity across the fracture, which could be achieved by assigning E_{prop} a much higher value such as 40 GPa, and it actually leads to a more uniform $|\sigma_{xx}|$ field (less $|\sigma_{xx}|$ perturbations around the fracture). As a result, the fracture conductivity loss would be higher, and the cumulative gas production would be lower.

Finally, we investigate the impact of permeability anisotropy, another type of anisotropy besides elastic anisotropy. We design the permeability tensor to be

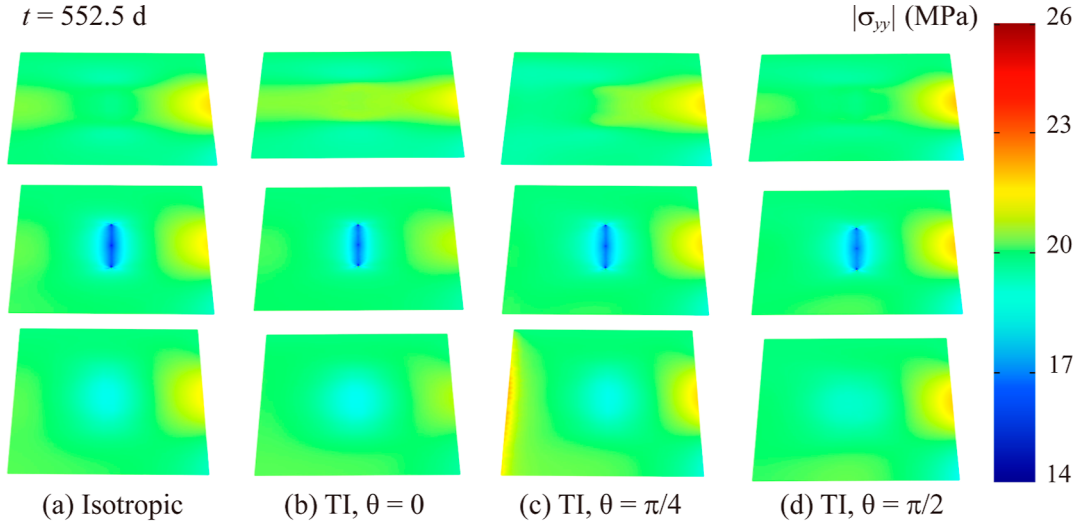


Figure 9. Comparisons of the total stress σ_{yy} field for isotropic and transversely isotropic shale. The profiles are shown for the top (z = 60 m), middle (z = 30 m), and bottom layers (z = 0 m), but we view them from a different direction compared with that in Figure 7.

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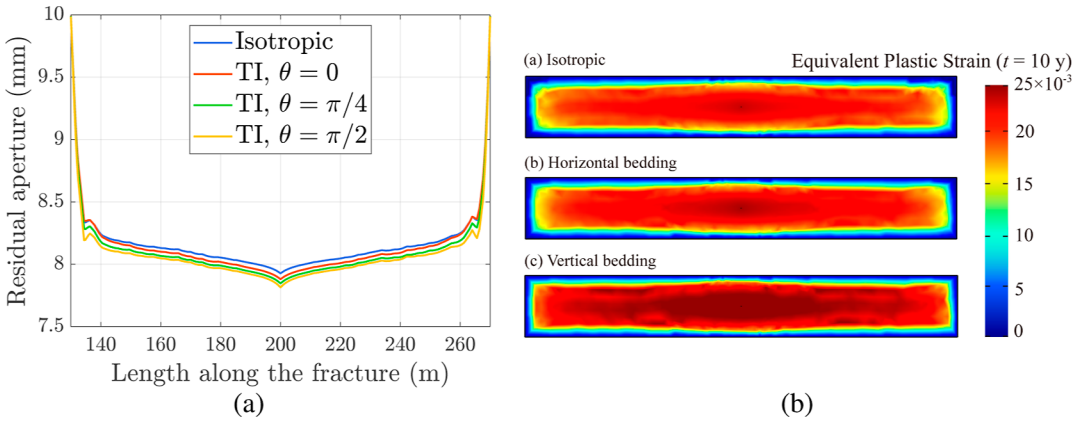


Figure 10. (a) Impact of anisotropy on the residual fracture aperture at $z = 30$ m and $t = 1000$ days. "TI" is abbreviated for "Transversely Isotropic". (b) Comparisons of the plastic strain for isotropic and transversely isotropic shale.

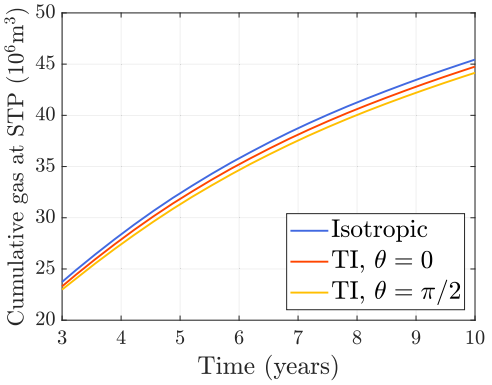


Figure 11. Impact of anisotropy on the cumulative gas production. "TI" is abbreviated for "Transversely Isotropic".

$$\boldsymbol{k} = \begin{bmatrix} 0.6 & 0 & 0.4 \\ 0 & 1 & 0 \\ 0.4 & 0 & 0.6 \end{bmatrix} k_0 \tag{56}$$

The comparison results with the isotropic permeability case are shown in Figures 12 and 13. Note that for both cases, the elastic anisotropy with $\theta = \pi/4$ is retained. We can see that an anisotropic permeability tensor $\boldsymbol{k}$ inhibits the gas production and exaggerates the deviation of the $|\sigma_{xx}|$ contour (on top) and gas pressure contour (on top and bottom) from the center line $x = 100$ m. Inspired by Figure 12, we deduce that the stress shadow mechanism will also be affected in this case.

6. CLOSURE

In the first part of this paper, we extend the approach proposed by Zhang and Borja[32] to consider micro-inhomogeneity and micro-anisotropy in a macroscopically anisotropic porous medium such as shale. The approach based on mixture theory adopts solid density ρ_s and porosity ϕ as variables, through which a complete set of anisotropic poroelastic constants is developed, and it also confirms the veracity of the intrinsic poroelastic constants proposed by Cheng[33] because the author chose the same variables (solid volume and porosity). While it is difficult to assess the micro-anisotropy effects using Cheng's approach,[33] our approach makes the first attempt toward anisotropy. Generalized versions of C_ϕ and C_ψ are developed, and bounds on material constants are also given. In the isotropic case, our approach could reproduce the same results as Cheng.[33]

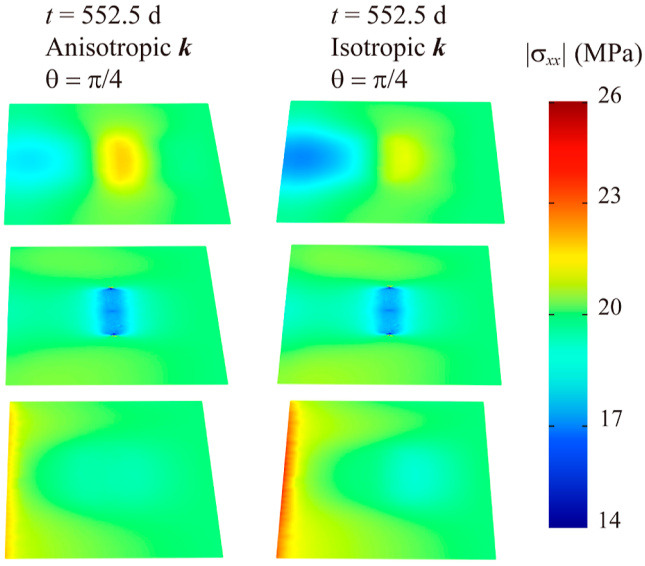


Figure 12. On the basis of elastic anisotropy introduced in Figure 7c, we add permeability anisotropy, which has the same plane of isotropy as $\mathbb{M}$,[31] and its impact is shown on the left and Figure 13. The profiles are shown for the top ($z = 60$ m), middle ($z = 30$ m), and bottom layers ($z = 0$ m), but we view them from a different direction compared with that in Figure 7.

In the second part of this paper, we examine the impact of anisotropy in shale gas reservoirs by assuming micro-homogeneity. During the gas production process, the total stress would exhibit either a U-shaped or an inverted U-shaped temporal variation around the discrete fracture, so a constant total stress assumption may not correctly capture the dynamics near the fracture. The formation anisotropy has a larger impact on σ_{xx} parallel to the wellbore than σ_{yy}, which subsequently affects the fracture deformation, fracture conductivity, and gas cumulative production. For inclined bedding planes, the formation and permeability anisotropy decrease the production rate and lead to the stress distribution reorientation. For future work, upon the determination of all the material constants proposed in the first part through laboratory experiments, a complete analysis considering micro-inhomogeneity will be performed. In addition, the double porosity nature and multiple fracture interactions are also important and might be left for a potential future investigation.

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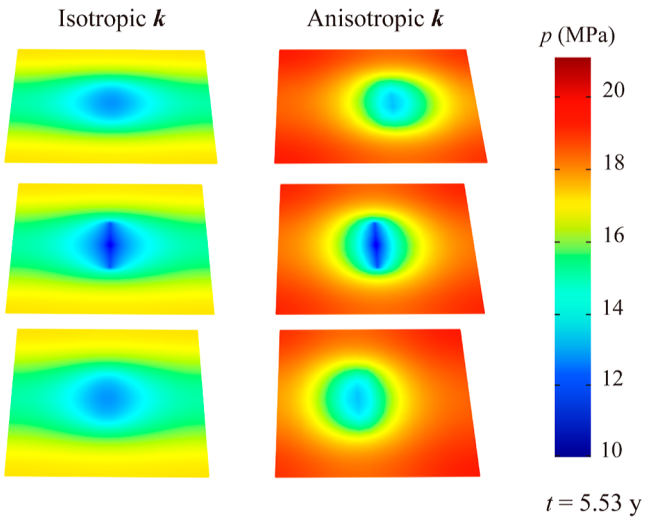


Figure 13. Impact of permeability anisotropy on the gas pressure field at 5.53 years. The profiles are shown for the top (z = 60 m), middle (z = 30 m), and bottom layers (z = 0 m), but we view them from a different direction compared with that in Figure 7.

■ ASSOCIATED CONTENT

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Notes

The authors declare no competing financial interest.

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■ NOMENCLATURE

Roman Letter

$\mathbf{1}$	second-order identity tensor
$\overline{\mathcal{B}},\mathcal{B},\partial\mathcal{B}$	closed domain, open domain, and boundary
$\partial\mathcal{B}_u,\partial\mathcal{B}_t$	essential and natural boundaries for $\boldsymbol{u}$
$\partial\mathcal{B}_p,\partial\mathcal{B}_q$	essential and natural boundaries for p
$\boldsymbol{b}$	prescribed body force vector
C^*	generalized drained compressibility, and its isotropic counterpart is $1/K$
C_s, $\overline{C}_s$, C_s''	reciprocal of K_s, $\overline{K}_s$, and K_s''
$C_s'^*$, C_s'	generalized unjacketed bulk compressibility and its isotropic counterpart
C_ϕ	drained porosity compressibility
C_ψ	unjacketed porosity compressibility
$\mathbb{C}_s'$	fourth-order unjacketed bulk compliance tensor
E, E_v, E_h	Young's modulus
$\boldsymbol{F}$	mass flux vector
G, G_{vh}	shear modulus
J	deformation Jacobian
K	drained bulk modulus by assuming isotropy
K^*	generalized drained bulk modulus, $K^* = (\mathbf{1}{:}\mathbb{M}{:}\mathbf{1})/9$. In general, K^* is not equal to $1/C^*$, except for the case of isotropy
K_f	fluid bulk modulus
$K_{m,i}$, $\overline{K}_m$	each mineral's bulk modulus and the dominant bulk modulus
K_s, $\overline{K}_s$	drained and effective bulk moduli of the composite solid
K_s'	unjacketed bulk modulus by assuming isotropy, reciprocal of C_s'
K_s''	unjacketed pore modulus
K_x	a new modulus term introduced in eq 6 considering the effect of p on ρ_s
K_ψ	reciprocal of C_ψ
k_0	initial absolute permeability of matrix
$\boldsymbol{k}$	permeability tensor
k_{F0}	initial absolute permeability of hydraulic fracture
$\mathbb{M}$,M_{ijkl}	fourth-order drained bulk stiffness tensor and its component. For detailed expressions, please refer to Section 4.1 in Zhang and Borja[32] or an in-depth tutorial by Nemeth.[37] Note in Zhang and Borja,[32] the authors adopted $\mathbb{C}^e$ for this stiffness tensor. Briefly speaking, for transversely isotropic elasticity with a given plane of isotropy characterized by $\boldsymbol{n}$, $\mathbb{M}$ is described by five material constants E_v, E_h, ν_{hh}, ν_{vh}, and G_{vh}; while for isotropic elasticity, $\mathbb{M}$ is described by two material constants E and ν
$\mathbb{M}^{-1}$	fourth-order drained bulk compliance tensor
M_g	gas molar mass
m	mass of adsorbed gas per solid volume
N	Biot modulus excluding the fluid compressibility term ϕ/K_f
$\boldsymbol{n}$	vector normal to the bedding plane
$\boldsymbol{n}_\partial$	outward normal vector defined on $\partial\mathcal{B}$
P	mean total compressive stress widely used in poroelasticity, equivalent to $-\sigma_t$
P'	Terzaghi effective compressive pressure, $P' = P - p$
P_L	Langmuir pressure
p_s, p	intrinsic solid pressure and fluid pressure
$p^{(I)}$, $p^{(II)}$	fluid pressure in drained (I) and unjacketed tests (II)

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$\boldsymbol{q}$	Darcy flux vector
R	universal gas constant
r_w	well radius
T	reservoir temperature
t	time, the units in the numerical example are second, day, and year
$\boldsymbol{u}$	displacement vector field
V, V_s, V_p	bulk volume, solid volume, and pore volume in a REV
V_L	Langmuir volume
v_i	each mineral's volume fraction
$\boldsymbol{v}_s$	solid skeleton velocity vector
w_{F0}	initial aperture of hydraulic fracture
Z	gas compression factor, $Z = 1$ for ideal gas

Greek Letter

$\boldsymbol{\alpha}$, α	Biot tensor and Biot coefficient scalar
$\boldsymbol{\varepsilon}$, ϵ_v	infinitesimal strain tensor $\varepsilon_{ij} = (u_{i,j} + u_{j,i})/2$ and volumetric strain $(=\Delta V/V = \varepsilon_{ii})$
$\boldsymbol{\varepsilon}^{(I)}$, $\boldsymbol{\varepsilon}^{(II)}$	strain tensor in drained (I) and unjacketed tests (II)
θ	bedding plane orientation depicted in Figure 3
μ_g	gas viscosity
ν, ν_{hh}, ν_{vh}	Poisson's ratio
$\boldsymbol{\xi}$, ξ_{ij}	arbitrary nonzero symmetric second-order tensor
ρ_g, ρ_{gstd}	gas density and its value at the standard condition
ρ_r	constant rock bulk density
ρ^s, ρ_s	partial and intrinsic mass densities of the solid $\rho^s = \rho_s(1-\phi)$
$\boldsymbol{\sigma}$, σ_t, σ_{xx}, σ_{yy}	total stress tensor, its mean value, and its components
$\boldsymbol{\sigma}^{(I)}$, $\boldsymbol{\sigma}^{(II)}$	total stress tensor in drained (I) and unjacketed tests (II)
$\boldsymbol{\sigma}''$	effective stress tensor
ϕ, ϕ^L, ϕ_0	Eulerian porosity, Lagrangian porosity, and the initial value

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